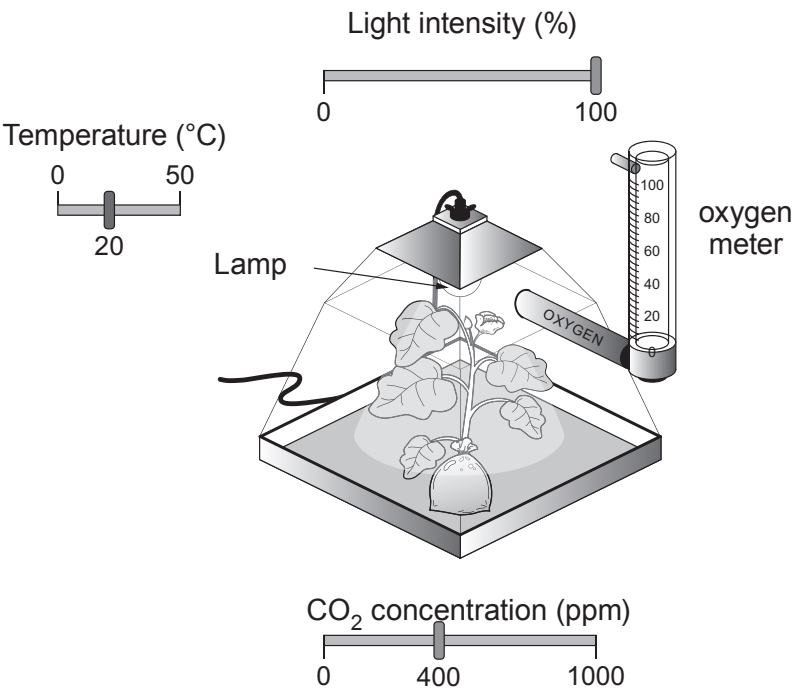


6. In a classroom exercise some students used a computer simulation to investigate the factors affecting the rate of photosynthesis. A well-watered plant, was placed inside a transparent sealed container with a high intensity lamp above the plant. The air in the airtight container was continually monitored in order to measure the rate of O₂ production
- The light intensity of the lamp could be varied between 0 - 100 %.
 - The CO₂ concentration of the air inside the sealed container could be varied between 0 – 1000 ppm.
 - The temperature inside the sealed container could be varied between 0 – 50 °C.



The table below shows the results obtained by one of the students.

Reading number	Light intensity (%)	Temperature (°C)	CO ₂ concentration (ppm)	O ₂ production (cm ³ /h)
1	0	10	0	0.0
2	20	10	200	3.1
3	40	10	200	3.1
4	40	10	400	3.1
5	40	20	400	34.7
6	60	20	400	41.7
7	80	20	400	41.7
8	100	25	400	41.7
9	100	25	600	47.3



Examiner
only

(a) Explain how oxygen production can be used as a measure of the rate of photosynthesis. [3]

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(b) Explain why oxygen production remains at 3.1 cm³/h for readings **2, 3 and 4**. [2]

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(c) Identify the limiting factor for readings **6, 7, 8 and 9** and explain your answer. [2]

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(d) State why the container must be sealed. [1]

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Examiner
only

- (e) Sian and Dafydd are students who carried out the computer simulation. Sian suggested that they try to set up a ‘live version’ of the apparatus in the laboratory. Dafydd said that if they did this, a problem could arise which would affect the validity of the experiment. Suggest the problem that may arise and how this could be solved. [2]

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